Lab 5

Submission by: **April 7th - 4pm**

1) [0.1 points]
Recall the following grammar:

$$
\begin{aligned}
E &\to E + T \mid T \\
T &\to T \times F \mid F \\
F &\to (E) \mid \mathtt{a}
\end{aligned}
$$

Give parse trees and derivations for each string.

a. `a`
b. `a+a`
c. `a+a+a`
d. `((a))`

2) [0.15 points]

Answer each part for the following context-free grammar G.

$$
\begin{aligned}
R &\to XRX \mid S \\
S &\to \mathtt{a}T\mathtt{b} \mid \mathtt{b}T\mathtt{a} \\
T &\to XTX \mid X \mid \varepsilon \\
X &\to \mathtt{a} \mid \mathtt{b}
\end{aligned}
$$

a. What are the variables of G?
b. What are the terminals of G?
c. Which is the start variable of G?
d. Give three strings in $L(G)$.
e. Give three strings *not* in $L(G)$.
f. True or False: $T \Rightarrow \mathtt{aba}$.
g. True or False: $T \overset{*}{\Rightarrow} \mathtt{aba}$.
h. True or False: $T \Rightarrow T$.
i. True or False: $T \overset{*}{\Rightarrow} T$.
j. True or False: $XXX \overset{*}{\Rightarrow} \mathtt{aba}$.
k. True or False: $X \overset{*}{\Rightarrow} \mathtt{aba}$.
l. True or False: $T \overset{*}{\Rightarrow} XX$.
m. True or False: $T \overset{*}{\Rightarrow} XXX$.
n. True or False: $S \overset{*}{\Rightarrow} \varepsilon$.
o. Give a description in English of $L(G)$.

3) [0.6 points]
Give context free grammars that generate the following languages. The alphabet is $\Sigma = \{0,1\}$.

a. $\{w \mid w$ contains at least three 1s$\}$
b. $\{w \mid w$ starts and ends with the same symbol$\}$
c. $\{w \mid$ the length of w is odd$\}$
d. $\{w \mid$ the length of w is odd and its middle symbol is a 0$\}$
e. $\{w \mid w = w^{\mathcal{R}}$, that is, w is a palindrome$\}$
f. The empty set

4) [0.5 points]
Give state diagrams of pushdown automata for the languages in exercise 3)

5) [0.4 points]
Give context-free grammars generating the following languages.

a. The set of strings over the alphabet $\{\mathtt{a},\mathtt{b}\}$ with more a's than b's
b. The complement of the language $\{\mathtt{a}^n\mathtt{b}^n \mid n \geq 0\}$

6) [0.25 points]
Convert the following CFG into an equivalent in Chomsky Normal Form using the procedure discussed at lecture.

$A \rightarrow BAB \mid B \mid \varepsilon$
$B \rightarrow 11 \mid \varepsilon$